

Causal ML: shrinkage and identification

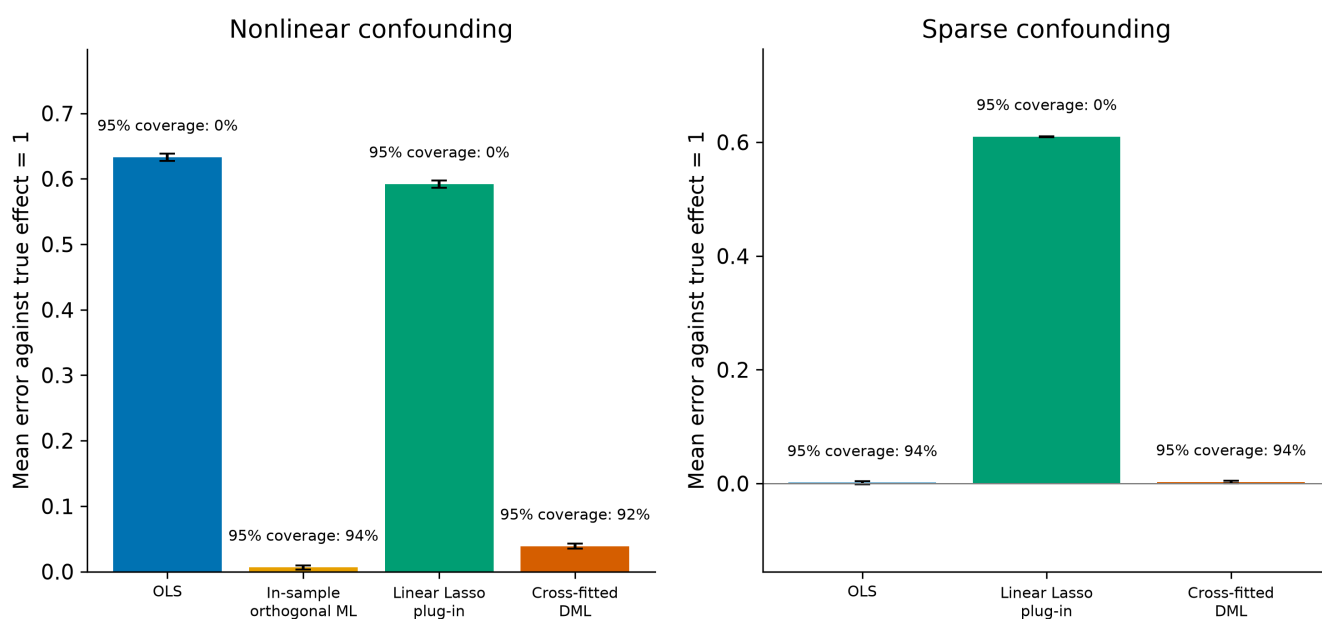
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Question

When can regularization distort an estimated treatment effect, and what can cross-fitting and sensitivity diagnostics tell us?

Method

We compare OLS, plug-in machine learning and cross-fitted double machine learning in two seeded designs with known treatment effects. We estimate 401(k) eligibility effects using partially linear and interactive models and examine propensity overlap and sensitivity to unobserved confounding. We compare staggered-adoption estimators and a late-cohort balanced event-study window using never-treated controls. We report Monte Carlo uncertainty, effect intervals and the assumptions required for causal interpretation.



1000 repetitions per design; bars: 95% Monte Carlo intervals for mean bias.
Coverage describes effect intervals. Nonlinear design: boosting nuisances; sparse design: Lasso nuisances.

Published 1991 SIPP extract via DoubleML; the did package's MPDTA county panel; seeded synthetic experiments.

Three findings for the decision-maker

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1. In 1,000 sparse-design simulations, linear Lasso plug-in bias is 0.610, versus 0.003 for cross-fitted DML; 95% interval coverage is 0.0% versus 93.8%.
2. Cross-fitting does not always improve the result: in the nonlinear design, in-sample orthogonal ML bias is 0.007, versus 0.039 for cross-fitted DML. These fixed-design comparisons do not establish a universal ranking.
3. In the historical SIPP sample, the RF-IRM estimate is \$8,172. The sensitivity robustness values are 8.41% for the effect bound and 4.94% for the confidence bound under the specified confounding model; they do not prove absence of confounding.

Limitations

- Designs and learner families differ; their rankings are not universal.
- IPW ESS measures weight concentration, not causal validity or propensity-estimation uncertainty.
- Historical eligibility is not randomized or modern policy participation.
- Balanced and full-sample event studies target different cohorts/windows.
- Group inference conditions on fitted nuisance models; transportability is untested.

Detailed assumptions, diagnostics and source lineage: the technical working paper and docs/PROVENANCE.md.